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global_data ={
  "attribute_1": "total_cost",
  "attribute_2": "material_continuity",
  "attribute_3": "electrical_conductivity",
  "attribute_4": "corrosion_resistance",
  "attribute_5": "weight",

  "direction_attribute_1": "negative",
  "direction_attribute_2": "positive",
  "direction_attribute_3": "positive",
  "direction_attribute_4": "positive",
  "direction_attribute_5": "negative",

  "attribute_1_min": 5,
  "attribute_1_max": 500,
  "attribute_2_min": 0,
  "attribute_2_max": 1,
  "attribute_3_min": 10,
  "attribute_3_max": 110,
  "attribute_4_min": 0,
  "attribute_4_max": 100,
  "attribute_5_min": 5,
  "attribute_5_max": 55,

  "copper_price_per_ton": 9500,
  "aluminum_price_per_ton": 2500,
  "copper_price_per_kg": 9.5,
  "aluminum_price_per_kg": 2.5
}
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regional_data=[
  {
    "region": "United States",
    "copper_product_market_share": 0.75,
    "aluminum_product_market_share": 0.25,
    "weight_attribute_1": 0.32,
    "weight_attribute_2": 0.10,
    "weight_attribute_3": 0.27,
    "weight_attribute_4": 0.18,
    "weight_attribute_5": 0.13
  },
  {
    "region": "China",
    "copper_product_market_share": 0.90,
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    "aluminum_product_market_share": 0.10,
    "weight_attribute_1": 0.47,
    "weight_attribute_2": 0.05,
    "weight_attribute_3": 0.29,
    "weight_attribute_4": 0.11,
    "weight_attribute_5": 0.08
  },
  {
    "region": "India",
    "copper_product_market_share": 0.80,
    "aluminum_product_market_share": 0.20,
    "weight_attribute_1": 0.37,
    "weight_attribute_2": 0.08,
    "weight_attribute_3": 0.23,
    "weight_attribute_4": 0.23,
    "weight_attribute_5": 0.09
  },
  {
    "region": "Europe",
    "copper_product_market_share": 0.70,
    "aluminum_product_market_share": 0.30,
    "weight_attribute_1": 0.22,
    "weight_attribute_2": 0.12,
    "weight_attribute_3": 0.26,
    "weight_attribute_4": 0.22,
    "weight_attribute_5": 0.18
  }
]

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product_data=[
  {
    "region": "United States",
    "dominant material": "cu",
    "product": "550W Monocrystalline TOPCon Solar PV Module",
    "copper_kg": 0.50,
    "aluminum_kg": 1.30,
    "total_mass_kg": 26.0,
    "non_material_cost_per_unit": 140.00,
    "attribute_2_value": 1,
    "attribute_3_value": 100,
    "attribute_4_value": 85,
    "attribute_5_value": 26.0
  },
  {

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"region": "United States",
"dominant material": "al",
"product": "550W Monocrystalline TOPCon Solar PV Module",
"copper_kg": 0.05,
"aluminum_kg": 1.55,
"total_mass_kg": 25.8,
"non_material_cost_per_unit": 140.00,
"attribute_2_value": 0,
"attribute_3_value": 61,
"attribute_4_value": 60,
"attribute_5_value": 25.8
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "550W Monocrystalline TOPCon Solar PV Module",
  "copper_kg": 0.50,
  "aluminum_kg": 1.30,
  "total_mass_kg": 26.0,
  "non_material_cost_per_unit": 37.00,
  "attribute_2_value": 1,
  "attribute_3_value": 100,
  "attribute_4_value": 85,
  "attribute_5_value": 26.0
},
{
  "region": "China",
  "dominant material": "al",
  "product": "550W Monocrystalline TOPCon Solar PV Module",
  "copper_kg": 0.05,
  "aluminum_kg": 1.55,
  "total_mass_kg": 25.8,
  "non_material_cost_per_unit": 37.00,
  "attribute_2_value": 0,
  "attribute_3_value": 61,
  "attribute_4_value": 60,
  "attribute_5_value": 25.8
},
{
  "region": "India",
  "dominant material": "cu",
  "product": "550W Monocrystalline TOPCon Solar PV Module",
  "copper_kg": 0.50,
  "aluminum_kg": 1.30,
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"total_mass_kg": 26.0,
"non_material_cost_per_unit": 100.00,
"attribute_2_value": 1,
"attribute_3_value": 100,
"attribute_4_value": 85,
"attribute_5_value": 26.0
},
{
  "region": "India",
  "dominant material": "al",
  "product": "550W Monocrystalline TOPCon Solar PV Module",
  "copper_kg": 0.05,
  "aluminum_kg": 1.55,
  "total_mass_kg": 25.8,
  "non_material_cost_per_unit": 100.00,
  "attribute_2_value": 0,
  "attribute_3_value": 61,
  "attribute_4_value": 60,
  "attribute_5_value": 25.8
},
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "550W Monocrystalline TOPCon Solar PV Module",
  "copper_kg": 0.50,
  "aluminum_kg": 1.30,
  "total_mass_kg": 26.0,
  "non_material_cost_per_unit": 75.00,
  "attribute_2_value": 1,
  "attribute_3_value": 100,
  "attribute_4_value": 85,
  "attribute_5_value": 26.0
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "550W Monocrystalline TOPCon Solar PV Module",
  "copper_kg": 0.05,
  "aluminum_kg": 1.55,
  "total_mass_kg": 25.8,
  "non_material_cost_per_unit": 75.00,
  "attribute_2_value": 0,
  "attribute_3_value": 61,
  "attribute_4_value": 60,
```

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    "attribute_5_value": 25.8
  }
]
```